

CTP0007 Issue H Calculation Procedure

This document describes the calculation procedure implemented in `coupling\_calculations.py` for the CTP0007 Issue H desktop application. Units are N, mm, MPa, and Nm unless otherwise noted. MPa is treated as N/mm².

1. Input Selection

The screw type and size select a screw record from the embedded CTP 0007 lookup data. For `SPECIAL` screws, the user-entered geometry is used where applicable.

Material yield strength is selected from the material database unless the editable `TYS MPa` field is changed by the user.

Tightening input priority is:

User-entered tightening torque.

User-entered percent of tensile yield strength, `% TYS`.

Standard tightening torque from the selected screw type, size, and material.

If both tightening torque and `% TYS` are entered, the calculation is rejected.

2. Geometry

Definitions:

`d` = thread major diameter

`p` = thread pitch

`ds` = shank diameter

`dg` = groove diameter

`dc` = contact diameter

`Sy` = bolt tensile yield strength

`n` = screw count

`PCD` = pitch circle diameter

Thread pitch diameter:

$$dp = d - 0.6495 * p$$

Thread root diameter:

$$dr = d - 1.2268 * p$$

Tensile stress diameter:

$$dt = (dp + dr) / 2$$

Tensile stress area:

$$As = \pi * ((dp + dr) / 4)^2$$

Shear and bending diameter:

$d_{sb} = d_t$ when shear plane is Thread

$d_{sb} = d_s$ when shear plane is Shank

Contact diameter:

d_c = special contact diameter for SPECIAL screws

d_c = special contact diameter when Nut contact is Special and value > 0

d_c = standard record contact diameter otherwise

Average annular nut/contact radius:

$R_o = d_c / 2$

$R_i = d_p / 2$

$R_n = (2 / 3) * ((R_o^3 - R_i^3) / (R_o^2 - R_i^2))$

The contact diameter must be greater than the pitch diameter.

3. Effective Lever Arm

Effective shear/bending lever arm:

$L_{eff} = 0.05$ mm for Stripper bolt

$L_{eff} = 0.15 * \text{pack_thickness_mm}$ for Drive bolt

$L_{eff} = 0.15 * \text{pack_thickness_mm}$ for Shim

Drive bolt and Shim require a positive pack thickness. Stripper bolt uses `N/A` pack thickness.

4. Tightening, Pretension, and Friction Torque

Friction definitions:

`mu\_sn` = screw/nut friction coefficient

`mu\_np` = nut/part friction coefficient

`mu\_pp` = part/part friction coefficient

`Tt` = selected tightening torque

`Fv` = axial pretension

Axial pretension from tightening torque:

$F_v = 1000 * T_t / (0.16 * p + 0.583 * \mu_{sn} * d_p + \mu_{np} * R_n)$

Axial pretension from `%TYS`:

$F_v = (\%TYS / 100) * A_s * S_y$

Reported tightening torque when `%TYS` is used:

$T_{t_reported} = (0.16 * p + 0.583 * \mu_{sn} * d_p + \mu_{np} * R_n) * F_v / 1000$

Preload percent of tensile yield strength:

$\text{preload_}\%TYS = F_v / (A_s * S_y) * 100$

Nut/part friction torque component:

$T_{np} = \mu_{np} * R_n * F_v / 1000$

Torsional tightening torque in the screw:

$$T_{\text{torsion}} = (0.16 * p + 0.583 * \mu_{\text{sn}} * dp) * F_v / 1000$$

Flange friction torque capacity:

$$T_{\text{friction}} = F_v * n * \mu_{\text{pp}} * \text{PCD} / 2 / 1000$$

Friction torque ratio is displayed as information:

$$\text{ratio}_{\%} = T_{\text{friction}} / T_{\text{duty}} * 100$$

Friction torque lower than duty torque is not a warning by itself. Any uncovered torque is carried forward into the bolt shear/bending calculation.

5. Duty Torque Cases

The app evaluates three duty torque cases:

Continuous

Peak

Momentary

For each case:

$$T_{\text{residual}} = \max(0, T_{\text{duty}} - T_{\text{friction}})$$

$$V_{\text{total}} = 2000 * T_{\text{residual}} / (\text{PCD} * n)$$

`V\_total` is the shear load per joint/bolt station.

If a sleeve is present and sleeve OD is greater than the thread major diameter:

$$V_{\text{sleeve}} = V_{\text{total}} * (\text{OD}^2 - d^2) / \text{OD}^2$$

$$V_{\text{bolt}} = V_{\text{total}} - V_{\text{sleeve}}$$

If no sleeve is present:

$$V_{\text{sleeve}} = 0$$

$$V_{\text{bolt}} = V_{\text{total}}$$

6. Sleeve Stress and Safety Factor

Sleeve check is only numeric when `sleeve OD > d`. Otherwise the result is `No Sleeve`.

Sleeve bending stress:

$$I_{\text{sleeve}} = \pi / 64 * (\text{OD}^4 - d^4)$$

$$\sigma_{\text{b_sleeve}} = V_{\text{sleeve}} * L_{\text{eff}} * \text{OD} / 2 / I_{\text{sleeve}}$$

Sleeve shear stress, as implemented:

$$\tau_{\text{sleeve}} = V_{\text{total}} / (\pi / 4 * \text{OD}^2)$$

Sleeve Von Mises stress:

$$\text{VM}_{\text{sleeve}} = \sqrt{\sigma_{\text{b_sleeve}}^2 + 3 * \tau_{\text{sleeve}}^2}$$

Sleeve yield safety factor:

$$\text{SF}_{\text{sleeve}} = \text{sleeve_yield} / \text{VM}_{\text{sleeve}}$$

Sleeve material options:

GMC 0401 = 245 MPa

GMC 0336 = 650 MPa

GMC0433 = 800 MPa

When sleeve OD is `N/A`, sleeve material and sleeve yield are `N/A`.

7. Bolt Shear/Bending Area Stress and Safety Factor

Bolt axial stress:

$$\sigma_{\text{axial}} = 4 * F_v / (\pi * d_{sb}^2)$$

Bolt bending stress:

$$I_{\text{bolt}} = \pi / 64 * d_{sb}^4$$

$$\sigma_{b_bolt} = V_{\text{bolt}} * L_{eff} * d_{sb} / 2 / I_{\text{bolt}}$$

Bolt shear stress:

$$\tau_{\text{bolt}} = V_{\text{bolt}} / (\pi / 4 * d_{sb}^2)$$

Bolt maximum Von Mises stress:

$$VM_{\text{bolt}} = \sqrt{(\sigma_{\text{axial}} + \sigma_{b_bolt})^2 + 3 * \tau_{\text{bolt}}^2}$$

Bolt yield safety factor:

$$SF_{\text{bolt}} = S_y / VM_{\text{bolt}}$$

The table column `Maxi Stress (VM) MPa` reports `VM\_bolt`.

8. Assembly Stress in Groove Area

If `dg <= 0`, the result is `No groove`.

Groove axial stress:

$$\sigma_{\text{groove}} = 4 * F_v / (\pi * d_g^2)$$

Groove twisting stress:

$$\tau_{\text{groove}} = 16 * T_{\text{torsion}} * 1000 / (\pi * d_g^3)$$

Groove Von Mises stress:

$$VM_{\text{groove}} = \sqrt{\sigma_{\text{groove}}^2 + 3 * \tau_{\text{groove}}^2}$$

Groove yield safety factor:

$$SF_{\text{groove}} = S_y / VM_{\text{groove}}$$

9. Assembly Stress in Thread Root Area

Thread-root axial stress:

$$\sigma_{\text{thread}} = F_v / A_s$$

Thread-root twisting stress:

$$\tau_{\text{thread}} = T_{\text{torsion}} * 1000 * d_r / 2 / (\pi / 32 * d_r^4)$$

Thread-root Von Mises stress:

$$\sigma_{\text{thread}} = \sqrt{\sigma_{\text{thread}}^2 + 3 * \tau_{\text{thread}}^2}$$

Thread-root yield safety factor:

$$SF_{\text{thread_root}} = S_y / \sigma_{\text{thread}}$$

10. Thread Pull-Out Stress

Thread engagement:

L = user-entered engagement

L = 1.2 * d when thread engagement mode is Thread/default

Thread pull-out area:

$$A_{\text{pullout}} = \pi / 2 * d_p * L$$

Engaged thread quantity:

$$N_{\text{engaged}} = L / p$$

First-thread load factor for `Emuge` or `Prevailing` screw/nut friction presets:

$$K = \max(1, 0.001762 * N_{\text{engaged}}^3 - 0.028314 * N_{\text{engaged}}^2 + 0.182016 * N_{\text{engaged}} + 0.720405)$$

First-thread load factor for other screw/nut friction presets:

$$K = \max(1, 0.002317 * N_{\text{engaged}}^3 - 0.039254 * N_{\text{engaged}}^2 + 0.416953 * N_{\text{engaged}} + 0.436158)$$

Thread pull-out stress:

$$\sigma_{\text{pullout}} = F_v / A_{\text{pullout}} * K$$

The tapped-hole yield strength field is optional. If entered, tapped-hole yield must be greater than or equal to thread pull-out stress.

11. Minimum Safety Factor

The reported minimum safety factor is the minimum of all numeric safety factors:

$$\text{minimum_SF} = \min(\text{all bolt shear/bending SF values, groove SF when groove exists, thread-root SF, all sleeve SF values when sleeve exists})$$

12. Checking Criteria

12.1 API 671 5th Edition

Yield safety factor limits for bolt shear/bending and sleeve area:

Continuous torque: SF $\geq$ 1.50

Peak torque: SF $\geq$ 1.15

Momentary torque: SF $\geq$ 1.00

12.2 API 671 4th Edition

Yield safety factor limits for bolt shear/bending and sleeve area:

Continuous torque: SF $\geq$ 1.25

Peak torque: SF $\geq$ 1.15

Momentary torque: SF $\geq$ 1.00

12.3 Groove Area

SF_groove $\geq$ 1.10

12.4 Thread Root Area

SF_thread_root $\geq$ 1.00 required

SF_thread_root $\geq$ 1.10 preferred

The preferred value is used because residual torque can reduce by about 20 percent upon loading.

12.5 Tightening Preload Percent

Stripper bolt: preload_%TYS $\leq$ 60

Shim: preload_%TYS $\leq$ 60

Drive bolt: preload_%TYS $\leq$ 75

12.6 Thread Pull-Out

tapped_hole_yield $\geq$ sigma_pullout

If tapped-hole yield is blank or zero, the app reports the pull-out stress with a note rather than a pass/fail result.

13. Input Validation

The calculation rejects invalid input before solving:

`PCD > 0`

`screw\_count  $\geq$  1`

`thread > 0`

`pitch > 0`

checking standard must be recognized

joint type must be recognized

pack thickness must be zero or positive

Drive bolt and Shim require positive pack thickness

contact diameter must be greater than zero

tightening torque and ``% TYS`` cannot both be entered

screw/nut, nut/part, and part/part friction coefficients must be greater than zero

sleeve OD must be zero or positive

tapped-hole yield must be zero or positive

lever arm must be zero or positive

14. Legacy Quick Friction Model Appendix

The module retains an older quick friction fastener model for compatibility and tests. It is not the main CTP0007 Issue H GUI workflow.

Effective friction radius:

$$r_{\text{eff}} = (\text{inner_radius} + \text{outer_radius}) / 2$$

when the radius model is uniform wear, otherwise:

$$r_{\text{eff}} = (2 / 3) * ((\text{outer_radius}^3 - \text{inner_radius}^3) / (\text{outer_radius}^2 - \text{inner_radius}^2))$$

Design torque:

$$T_{\text{steady}} = \text{transmitted_torque} * \text{service_factor}$$

$$T_{\text{design}} = \max(T_{\text{steady}}, \text{cyclic_torque}, \text{transient_torque})$$

Residual pretension after preload loss:

$$F_{\text{residual}} = F_{\text{initial}} * (1 - \text{preload_loss_percent} / 100)$$

Slip torque capacity:

$$T_{\text{slip}} = \mu * (F_{\text{residual}} * \text{bolt_count}) * r_{\text{eff}} * \text{friction_interfaces} / 1000$$

$$SF_{\text{slip}} = T_{\text{slip}} / T_{\text{design}}$$

Required clamp load:

$$F_{\text{required_total}} = T_{\text{design}} * 1000 / (\mu * r_{\text{eff}} * \text{friction_interfaces})$$

$$F_{\text{required_per_bolt}} = F_{\text{required_total}} / \text{bolt_count}$$

$$F_{\text{required_initial}} = F_{\text{required_per_bolt}} / (1 - \text{preload_loss_percent} / 100)$$

Bolt yield and proof loads:

$$F_{\text{yield}} = \text{yield_strength} * \text{tensile_area}$$

$$F_{\text{proof}} = \text{proof_strength} * \text{tensile_area}$$

Bolt/joint stiffness split:

$$\phi = k_{\text{bolt}} / (k_{\text{bolt}} + k_{\text{joint}})$$

$$F_{\text{bolt_service}} = F_{\text{residual}} + \phi * F_{\text{axial}}$$

$$F_{\text{residual_service}} = F_{\text{residual}} - (1 - \phi) * F_{\text{axial}}$$

Yield utilization values:

$$\text{assembly_yield_utilization} = F_{\text{initial}} / F_{\text{yield}}$$

$$\text{service_yield_utilization} = F_{\text{bolt_service}} / F_{\text{yield}}$$

$$\text{required_preload_yield_utilization} = F_{\text{required_initial}} / F_{\text{yield}}$$

Legacy quick model warnings include low service factor, transient/cyclic torque governing, slip capacity below demand, preload/yield utilization above the selected limit, preload above proof load, and joint separation.

15. Spreadsheet Cell Cross-Reference

The following table cross-references the procedure formulas to the source CTP0007 spreadsheet cells. References are from the template workbook: C:\Users\jtpkyyu\Smiths Group\Engineering Couplings - Coupling Technical Programs (CTP)\CTP 0007.xltx

Where the app intentionally extends the spreadsheet, the table identifies the original source cell for the calculated value and notes the app-specific control or criterion.

Procedure topic	Formula/result in this procedure	CTP0007 spreadsheet cells	Notes
Reference workbook	Cross-reference source for the procedure	Workbook: C:\Users\jtpkyyu\Smiths Group\Engineering Couplings - Coupling Technical Programs (CTP)\CTP 0007.xltx	Template sheets used: CTP 0007, Donn'es Vis, CTM 33 30 10, Bolt Bending, Sleeve Bending.
Screw type and size lookup	Selected Type and Size drive default bolt geometry and standard torque	CTP 0007!A14, E14; Donn'es Vis!B3:X168; formulas CTP 0007!U22, U23, U30, U31, U35, U37	The app database is seeded from the hidden Donn'es Vis screw rows and lets SPECIAL geometry be entered manually.
Basic joint inputs	PCD, screw count, material, and optional manual TYS	CTP 0007!I14, N14, R14, X14	The app labels X14 as editable TYS MPa; if blank, material yield comes from the database.
Friction coefficients	mu_sn, mu_np, mu_pp from presets or custom values	CTP 0007!A19, D19, G19; preset selectors A20, D20, G20; custom AD2:AF2; Donn'es Vis!Z34:AA49	Directly corresponds to the three friction fields in the app.
Standard tightening torque	Use selected screw type, size, and material standard torque when enabled	CTP 0007!U37; Donn'es Vis!O3:X160; CTM 33 30 10!B15, B18	CTM 33 30 10 is the reference standard torque sheet; the app uses the Donn'es Vis torque lookup for the selected CTP screw row.
Thread geometry	$dp = d - 0.6495p$; $dr = d - 1.2268p$; $dt = (dp + dr) / 2$; $As = \pi((dp + dr)/4)^2$	CTP 0007!U22, U23, U24, U25, U26, U27; areas AB24, AB25	These are the main screw-data geometry rows.
Shear/bending diameter	dsb = dt for Thread shear plane, or shank diameter for Shank shear plane	CTP 0007!O28, U28, AB28; shank lookup through Donn'es Vis!B3:I168	The app uses the same Thread/Shank selection concept.
Groove diameter	No groove when groove diameter is blank/N/A/zero; otherwise use specified diameter	CTP 0007!O7, U30, AB30	In the spreadsheet U30 equal to zero drives No groove behavior in rows U67:U70.
Nut contact diameter and average nut radius	dc from standard contact or special contact; $Rn = (2/3)((Ro^3 - Ri^3)/(Ro^2 - Ri^2))$	CTP 0007!J31, P31, U31, U32, AB32	J31=Standard and P31=special input control U31. The app exposes Nut contact and Special contact dia beside Standard contact dia.
Sleeve geometry and sleeve yield	Sleeve check active only when sleeve OD exceeds bolt major diameter	CTP 0007!U33, O33, AB33, U34, AB34	The app adds sleeve material presets; the spreadsheet has the sleeve yield input at U34.
Effective lever arm	$Leff = 0.05$ for Stripper bolt; $Leff = 0.15$ pack thickness for Drive bolt or Shim	CTP 0007!U29	The spreadsheet has a direct Leff input. The app derives Leff from joint type and pack thickness per the requested workflow.
Duty torque cases	Continuous, Peak, Momentary duty torque cases	CTP 0007!J19, O19, R19	The template calculates O19 = 2 x J19 and R19 = 1.15 x O19. The app allows all three duty torque cases to be entered directly.
Tightening input and axial pretension	$Fv = 1000 Tt / (0.16p + 0.583 \mu_{sn} dp + \mu_{np} Rn)$, or $Fv = \%TYS As Sy$	CTP 0007!V19, Z19, U36; reported torque U38; preload ratio AD24; message A38	The app rejects simultaneous torque and %TYS input, then reports the implied value of the other basis.
Nut friction and torsional tightening torque	$T_{np} = \mu_{np} Rn Fv / 1000$; $T_{torsion} = (0.16p + 0.583 \mu_{sn} dp) Fv / 1000$	CTP 0007!U39, U40; labels A39:A40	U39 is the nut/contact friction torque component. U40 is the torsional tightening torque used in groove and thread-root checks.
Flange friction torque and ratio	$T_{friction} = Fv n \mu_{pp} PCD / 2 / 1000$; ratio = $T_{friction} / T_{duty}$	CTP 0007!P43, P44, S44, W44	The app displays the ratio as a percentage and does not warn solely because friction torque is below duty torque.
Residual torque and shear load	$T_{residual} = \max(0, T_{duty} - T_{friction})$; $V_{total} = 2000 T_{residual} / (PCD n)$	CTP 0007!P45:S45:W45 and P46:S46:W46	The app carries residual torque into the bolt/sleeve stress calculations.
Sleeve stress and safety factor	Sleeve shear split, bending stress, shear stress, VM stress, and sleeve SF	CTP 0007!P49:S49:W49, P50:S50:W50, P51:S51:W51, P52:S52:W52, P53:S53:W53; Sleeve Bending!B28, B30, B32	The GUI table follows the CTP 0007 row 48:53 direct sleeve stress block; Sleeve Bending supports the template allowable peak calculation.
Bolt shear/bending stress and safety factor	Bolt share, axial stress, bending stress, shear stress, VM stress, and bolt SF	CTP 0007!P56:S56:W56, P57:S57:W57, P61:S61:W61, P62:S62:W62, P63:S63:W63, P64:S64:W64; Bolt Bending!B32, B34, B36 and J49:J59	The app column Maxi Stress (VM) MPa corresponds to CTP 0007!P63/S63/W63 for the three duty cases.
Partial sleeve/shank block	Additional partial-sleeve shank shear block in the spreadsheet	CTP 0007!P58:S58:W58, P59:S59:W59, P60:S60:W60	The app reports the main bolt shear/bending SF from P64/S64/W64 equivalent calculations.
Groove assembly stress	sigma_groove, tau_groove, VM_groove, SF_groove	CTP 0007!U67, U68, U69, U70	Blank/N/A groove in the app maps to the spreadsheet No groove branch.
Thread-root assembly stress	sigma_thread, tau_thread, VM_thread, SF_thread_root	CTP 0007!U73, U74, U75, U76	The app reports the thread-root safety factor and warning state from this same calculation group.
Thread engagement and pull-out stress	$L = 1.2d$ by default or user-entered manual engagement; $A_{pullout} = \pi/2 dp L$; first-thread factor; pull-out stress	CTP 0007!U79, U80, U81, U82	The spreadsheet default is U79 = 1.2 x U22. The app also allows manual thread engagement and optional tapped-hole yield comparison.
Minimum safety factor and check text	Minimum of active numeric safety factors and warning text	CTP 0007!AB19; groups P53:S53:W53, P64:S64:W64, U70, U76	The spreadsheet AB19 formula contains the legacy check string. The app computes individual warnings in code so the selected criteria can be changed.

Procedure topic	Formula/result in this procedure	CTP0007 spreadsheet cells	Notes
Legacy spreadsheet check thresholds	Template check formula thresholds	CTP 0007!AB19	The template formula checks sleeve/bolt continuous <1.50, peak <1.25, momentary <1.10, groove <1.10, and thread-root <1.10.
API671 criteria selector	API671 5th: continuous >=1.50, peak >=1.15, momentary >=1.00. API671 4th: continuous >=1.25, peak >=1.15, momentary >=1.00.	No direct original CTP 0007 cell; implemented in app code	This is an app extension requested for the Issue H workflow, using the spreadsheet stress outputs as the checked values.
Tightening preload criteria	Stripper bolt and Shim <=60% TYS; Drive bolt <=75% TYS	CTM 33 30 10!B17 for the 75% reference utilization; no direct original CTP 0007 cell for the 60% app limit	The app applies the requested joint-type-specific maximum preload rule.
Tapped-hole yield check	Tapped-hole yield must be greater than or equal to thread pull-out stress when entered	Thread pull-out stress: CTP 0007!U82; no original tapped-hole yield input cell	The tapped-hole yield field and pass/fail warning are app additions.
Allowable peak reference	Allowable peak torque from bolt and sleeve detailed bending sheets	CTP 0007!AD4, AE4; Bolt Bending!B36; Sleeve Bending!B32	This remains a useful template cross-check. The app focuses on the three duty-case stress/SF results.